E-Beam Lithography

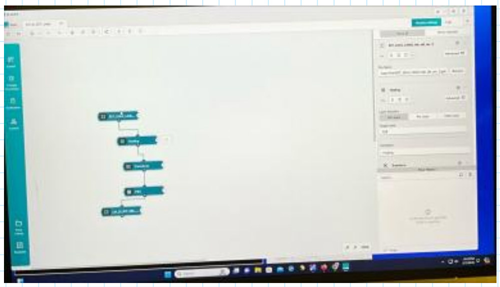
Friday, April 4, 2025 1:39 PM

https://cns1.rc.fas.harvard.edu/facilities/docs/2010-07-09%20EBL%20talk.pdf

Steps :
- GDS Marks
- Beamer
- Wecas
- SEM
- Wecas

GDS Marks → Beamer

1.) Get your GDS file with marks (layers)
2.) Edit your GDS in Klayout editormode to get the desired Layer for writing.
3.) Save it
4.) open Beamer App
5.) Create Process flow / Load existing flow → For IBTs There is already a flow file in the Pc with Beamer app
   - Load GDS
   - Healing
   - Transform
   - PEC ⇒ use right simulation
   - Select dose rate in the last process
   - save .con file



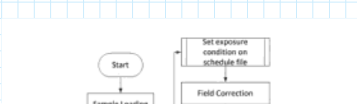
6.) Go to WECAS in test pc
   - Edit Schedule
   - Load .con file
   - Check estimated time
   - Dose rate calculation
   - Get estimate and Book tool
7.) Get the Spin coated Sample
8.) Load the Sample
   - Check exchange position in SEM app
   - On the machine check if sample stage in "LC"

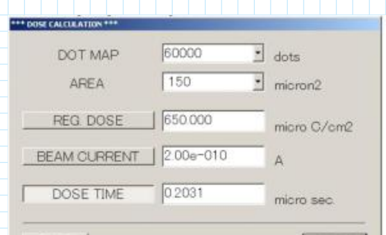
   - Open and Load the Sample
   - close and press "Evac". and wait till pressure Vacuum
   - open gate
   - Push rod in and unlock the stage and pull back
   - Close gate
9.) SEM app
   - click to Recall Beam
   - Eichenfield 3nA 120 ap
   - Isolation valve Button click
   - Faraday Cup
   - Zoom out (Mag 1) and center the green lines to center the circle
   - Press Measure Current
   - For 3nA if it not around 3, click adjust to increase or decrease the current.
   - Then go to Reference position
   - Find a dust and focus
   - Move stage often to avoid Charging
   - find gold particles
   - Focus and Stigma ⟷ alternatively BLK use
   - After seeing the particles clear
   - Move faraday Cup and Measure current / adjust
   - Then check Reference again for focus and clarity
10.) WECAS app
   - Edit Schedule
   - Load .Con file and Next
   - More pattern

Move stage to left bottom corner before measuring inclination

   - Measure Sample inclination and Next (the laser displacement meter should be 0.000_)
   - Field Correction (OK)
   - Calc.
   - Exposure
   - Wait till a few field is written to check for inclination error.

Look at the z preset value (maybe change it if you get the laser displacement meter not being in 0.0000





*** DOSE CALCULATION ***

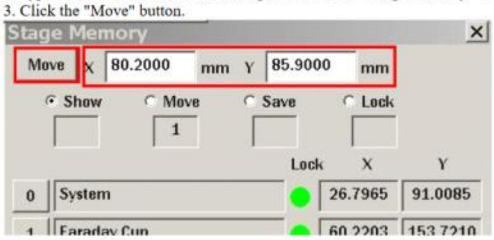
DOT MAP	60000	dots
AREA	150	micron2
REG. DOSE	650.000	micro C/cm2
BEAM CURRENT	2.00e-010	A
DOSE TIME	0.2031	micro sec.
CALC		EXIT

Flow chart of Elionix operating procedure

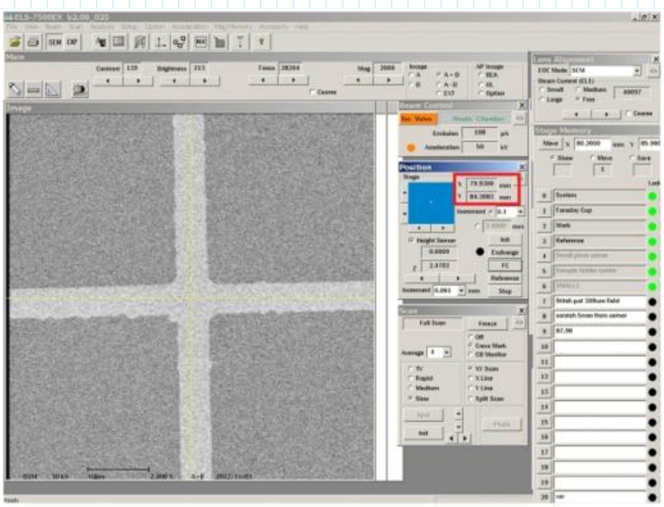
https://www.seas.upenn.edu/~nanosop/Elionix_Alignment_SOP.htm

6. Confirm the coordinates of Reg-A and Reg-B marks using SEM

1. Open the "Stage Memory" window by clicking the "Stage Memory" icon on the Tool bar.
2. Type the estimated coordinate of Reg-A mark in the "Stage Memory" window.
3. Click the "Move" button.



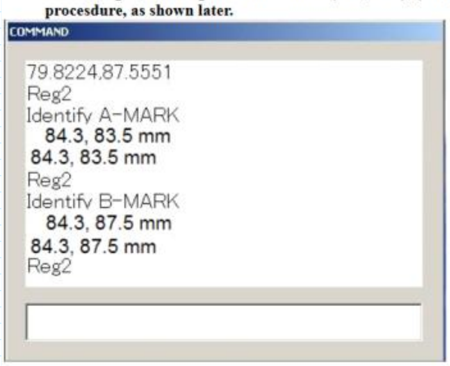
4. After the stage moves, open the beam shutter.
5. Find the Reg-A mark on the SEM image.
6. Check the "Cross Mark" in the "Scan" window, if not.
7. Move the center of the Reg-A mark to the center of the cross mark, using the mouse (select "Move Stage (Center)" from the drop down menu).
8. Increase the magnification (x2000 - x100000), and then move the center of the Reg-A mark to the center of the cross mark.
9. Record the coordinate of the Reg-A mark shown on the "Position" window.
   - This is the accurate coordinate of the Reg-A mark.
   - Ensure that the slow mode in the "Scan" window is selected.



10. Close the beam shutter.
11. Move the stage to the estimated coordinate of Reg-B mark in the same manner as above.
12. After the stage moves, open the beam shutter.
13. Move the center of the Reg-B mark to the center of the cross mark, and record the coordinate of the Reg-B mark in the same manner as above.
    - This is the accurate coordinate of the Reg-B mark.
14. Close the beam shutter.

IMPORTANT:

- **For example, it is assumed that the designed coordinates of the center, Reg-A, and Reg-B are (0,0), (0,-2), and (0,2) in mm, respectively. The sample size is 15 mm x 15 mm. Then, the absolute coordinates of the center, Reg-A, and Reg-B on the tool are (84.3, 85.5), (84.3, 83.5), and (84.3, 87.5) in mm, respectively. The relative relationship must be precise. The Elionix will adjust the real coordinates during exposure procesdure, as shown later.**



https://www.yumpu.com/en/document/read/19278682/exposure-procedure/18